

# AASD 4010

# Deep Learning - I

Applied AI Solutions Developer Program



# Module 9

# Transfer Learning

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# Agenda



Transfer Learning  
ULMFit using Fastai  
GPT-2  
GPT-3  
GPT-Neo  
BERT



# Transfer Learning

Transfer Learning is a process by which the learnings from a big pre-trained model is transferred to a custom problem in hand

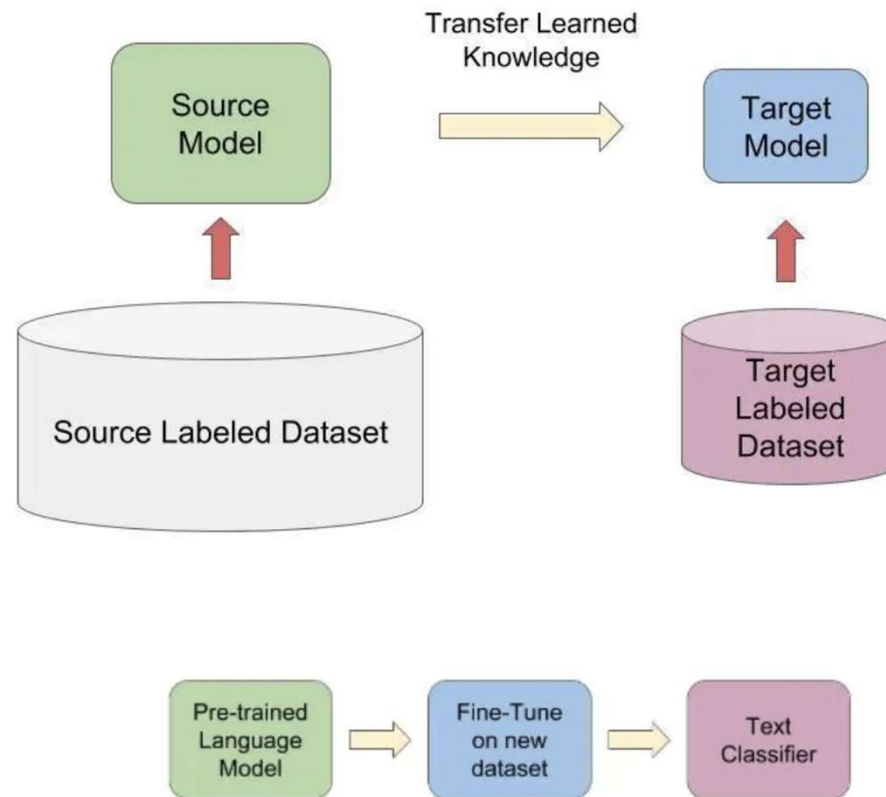
This eliminates the need to train a new model from scratch

Can leverage most of the learnings that happened in the first layers (strong low-mid level features)

This facilitates the transferring of the learnings (low, mid and some high-level features) from the big model thereby reducing the training (finetuning) time



# Transfer Learning



# ULMFiT



# ULMFiT

- Create a Base Language Model
- Finetune Base Language model on Task Specific Data
  - **Discriminative Fine-tuning** (Different learning rates for different layers)
  - **Slanted triangular learning rates** (High learning rates at start and low learning rates at later stages)
- Finetune Base Language model + Task Specific layers on Task specific data
  - **Gradual Unfreezing**



# Finetune Base LM on Task specific data

- Finetune Base Language model on Task Specific Data
  - **Discriminative Fine-tuning** (Different learning rates for different layers)

```
# Learning with Discriminative fine tuning
learn.fit_one_cycle(1, 1e-2)
learn.unfreeze()
learn.fit_one_cycle(1, 1e-3)
# Save encoder Object
learn.save_encoder('ft_enc')
```

# Finetune Base LM + Task specific layers on Task specific data

- **Gradual Unfreezing**

- Unfreeze last layer and finetune all unfrozen layers
- Unfreeze the next to last layer and repeat until all layers

```
# Creating Classifier Object
learn = text_classifier_learner(data_clas, AWD_LSTM, drop_mult=0.5)
# Add weights of finetuned Language model
learn.load_encoder('ft_enc')
# Fitting Classifier Object
learn.fit_one_cycle(1, 1e-2)
# Fitting Classifier Object after freezing all but last 2 layers
learn.freeze_to(-2)
learn.fit_one_cycle(1, slice(5e-3/2., 5e-3))
# Fitting Classifier Object - discriminative learning
learn.unfreeze()
learn.fit_one_cycle(1, slice(2e-3/100, 2e-3))
```

# Finetuning Steps

- Step 1: Download the dataset and Prepare the dataset
- Step 2: Build a Language Model data
- Step 3: Build a Classifier Model data
- Step 4: Fine-tune a pre-trained language model (ULMFiT - AWD\_LSTM) using `language_model_learner`
- Step 5: Use One-cycle policy to converge faster
- Step 6: Save the encoder
- Step 7: Build a classifier with fine-tuned saved encoder using `text_classifier_learner`
- Step 8: Fit the model using One-cycle policy



# ULMFit using Fastai

Kaggle Notebook



# Generative Pretrained Transformer

GPT



# Language Model

- A language model learns to predict the probability of a sequence of words
  - Predict  $p(w | h)$  – what is the probability of seeing the word  $w$  given a history of previous words  $h$  – where the history contains  $n-1$  words
- Language Models
  - **Statistical Language Models**
    - N-grams, Hidden Markov Models, CRF
  - **Neural Language Models**
    - Word2Vec, Glove, BERT, ELMO, and so on



# Generative Pretrained Transformer

GPT-2



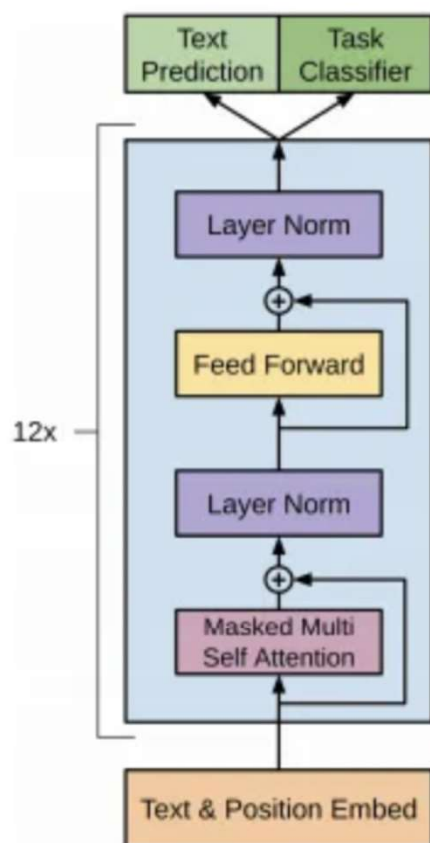
# GPT-2

- **1.5 billion** paramters
- Trained on **8 million** web pages from Reddit
- Applications
  - Text Generation
  - Machine Translation
  - Question Answering Systems

Can you please come **here** ?

The diagram shows the sentence "Can you please come here ?". A horizontal bracket is placed under the words "Can you please come". An upward-pointing arrow from this bracket is labeled "History". Another upward-pointing arrow points to the word "here", which is highlighted in red. Below this second arrow is the text "Word being predicted".

# GPT-2



| Parameters | Layers | $d_{model}$ |
|------------|--------|-------------|
| 117M       | 12     | 768         |
| 345M       | 24     | 1024        |
| 762M       | 36     | 1280        |
| 1542M      | 48     | 1600        |

Published

Not Published

<https://openai.com/blog/gpt-2-6-month-follow-up/>  
<https://openai.com/blog/gpt-2-1-5b-release/>

# Generative Pretrained Transformer 3

GPT-3



# GPT-3

- **175 billion** paramters
- Trained on **Common Crawl Dataset**

| Model Name            | $n_{\text{params}}$ | $n_{\text{layers}}$ | $d_{\text{model}}$ | $n_{\text{heads}}$ | $d_{\text{head}}$ | Batch Size | Learning Rate        |
|-----------------------|---------------------|---------------------|--------------------|--------------------|-------------------|------------|----------------------|
| GPT-3 Small           | 125M                | 12                  | 768                | 12                 | 64                | 0.5M       | $6.0 \times 10^{-4}$ |
| GPT-3 Medium          | 350M                | 24                  | 1024               | 16                 | 64                | 0.5M       | $3.0 \times 10^{-4}$ |
| GPT-3 Large           | 760M                | 24                  | 1536               | 16                 | 96                | 0.5M       | $2.5 \times 10^{-4}$ |
| GPT-3 XL              | 1.3B                | 24                  | 2048               | 24                 | 128               | 1M         | $2.0 \times 10^{-4}$ |
| GPT-3 2.7B            | 2.7B                | 32                  | 2560               | 32                 | 80                | 1M         | $1.6 \times 10^{-4}$ |
| GPT-3 6.7B            | 6.7B                | 32                  | 4096               | 32                 | 128               | 2M         | $1.2 \times 10^{-4}$ |
| GPT-3 13B             | 13.0B               | 40                  | 5140               | 40                 | 128               | 2M         | $1.0 \times 10^{-4}$ |
| GPT-3 175B or “GPT-3” | 175.0B              | 96                  | 12288              | 96                 | 128               | 3.2M       | $0.6 \times 10^{-4}$ |

# GPT-3

## Fine-tuning

The model is trained via repeated gradient updates using a large corpus of example tasks.



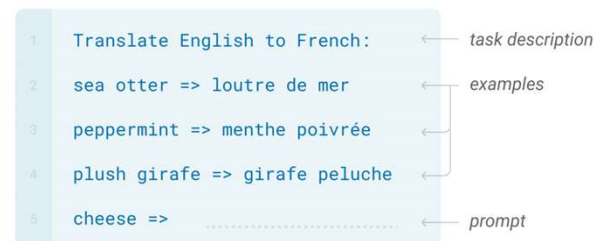
## Zero-shot

The model predicts the answer given only a natural language description of the task. No gradient updates are performed.



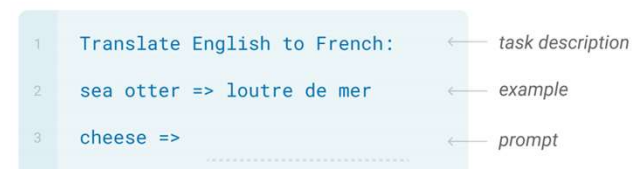
## Few-shot

In addition to the task description, the model sees a few examples of the task. No gradient updates are performed.



## One-shot

In addition to the task description, the model sees a single example of the task. No gradient updates are performed.



Paper: <https://arxiv.org/pdf/2005.14165.pdf>

Applications: <https://openai.com/blog/gpt-3-apps/>

API Pricing: <https://beta.openai.com/pricing>

# Generative Pretrained Transformer Neo

GPT-Neo



# GPT-Neo

Poor man's GPT-3 by <https://www.eleuther.ai/>

GPT-3 Vs GPT-Neo

<https://medium.com/georgian-impact-blog/gpt-neo-vs-gpt-3-are-commercialized-nlp-models-really-that-much-better-f4c73ffce10b>



# Bidirectional Encoder Representations from Transformers

BERT



# BERT

BERT is powered by the following things

- Self-Attention
- Transformer

BERT is not powered by

- RNN

It is only 12 layers of Encoder Stack alone and no Decoder Stack



# Finetuning BERT

BERT



# Task: Classify positive or negative review



“a visually stunning rumination on love”

Reviewer #1

That’s a **positive** thing to say

“reassembled from the cutting room floor of any given daytime soap”

Reviewer #2

That’s **negative**

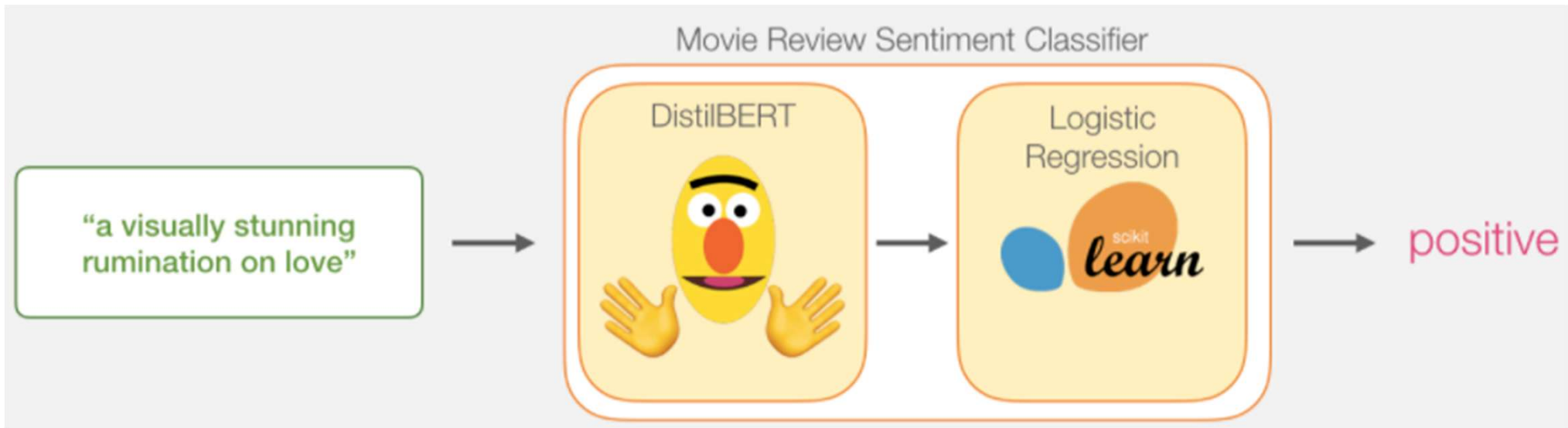
# Reviews dataset

| sentence                                                                                                | label |
|---------------------------------------------------------------------------------------------------------|-------|
| a stirring , funny and finally transporting re imagining of beauty and the beast and 1930s horror films | 1     |
| apparently reassembled from the cutting room floor of any given daytime soap                            | 0     |
| they presume their audience won't sit still for a sociology lesson                                      | 0     |
| this is a visually stunning rumination on love , memory , history and the war between art and commerce  | 1     |
| jonathan parker 's bartleby should have been the be all end all of the modern office anomie films       | 1     |

# Movie Review Classifier



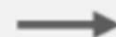
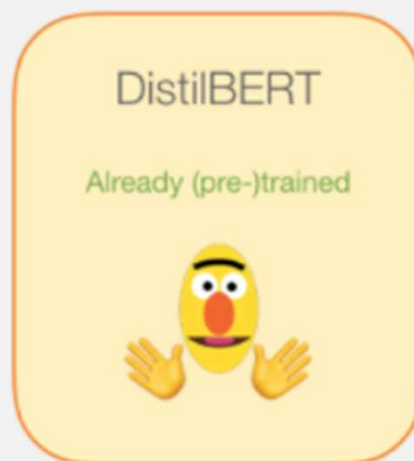
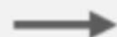
# Pretrained model + Linear Classifier



# Pretrained model

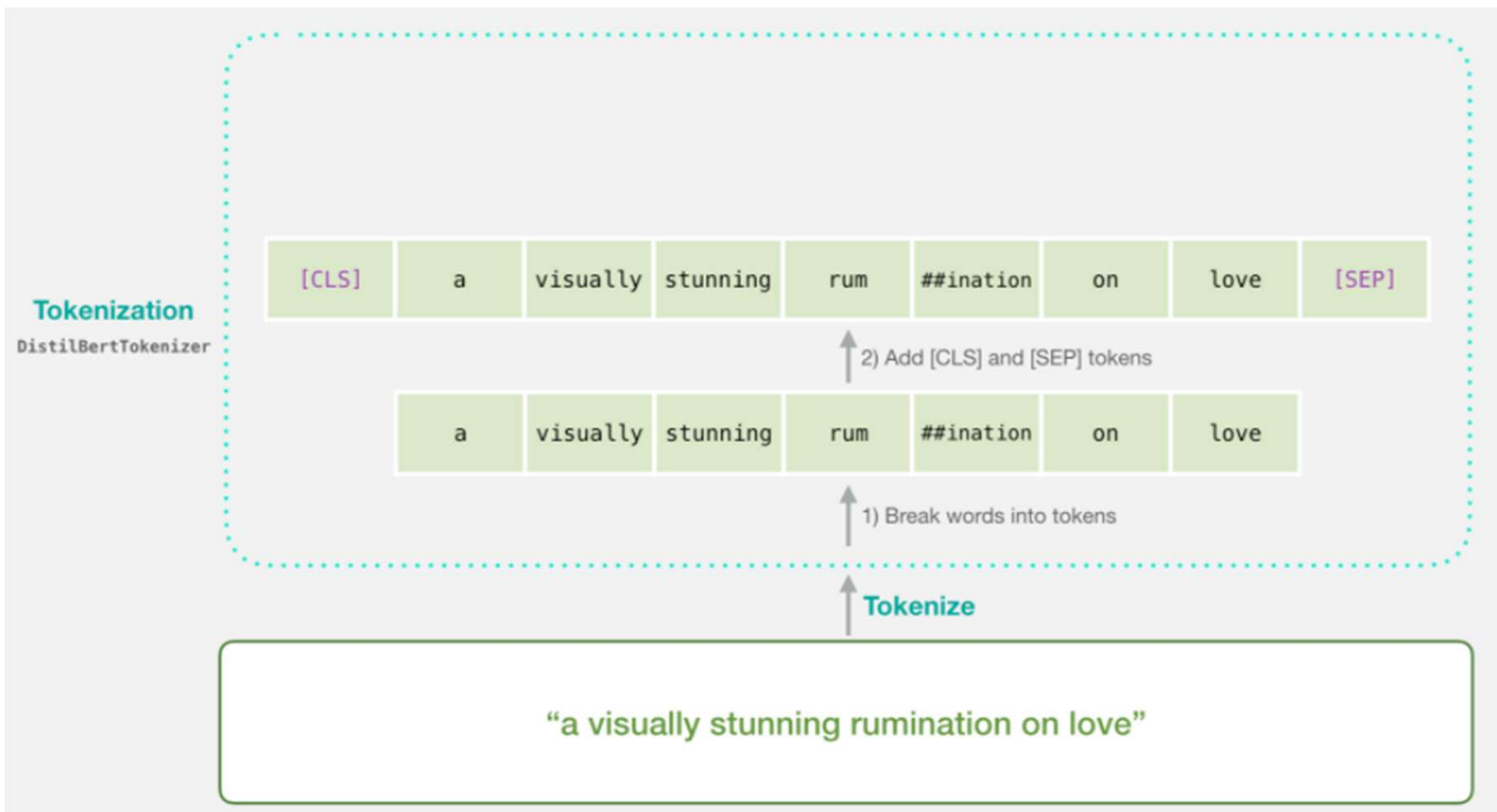
Step #1: Use DistilBERT to embed all the sentences

|       | Sentence                                                                                   | label |
|-------|--------------------------------------------------------------------------------------------|-------|
| 0     | a stirring , funny and finally transporting re imagining of beauty and the beast and 1930s | 1     |
| 1     | apparently reassembled from the cutting room floor of any given daytime soap               | 0     |
| ...   | ...                                                                                        | ...   |
| 1,999 | the movie is undone by a filmmaking methodology that 's just experimental enough           | 1     |

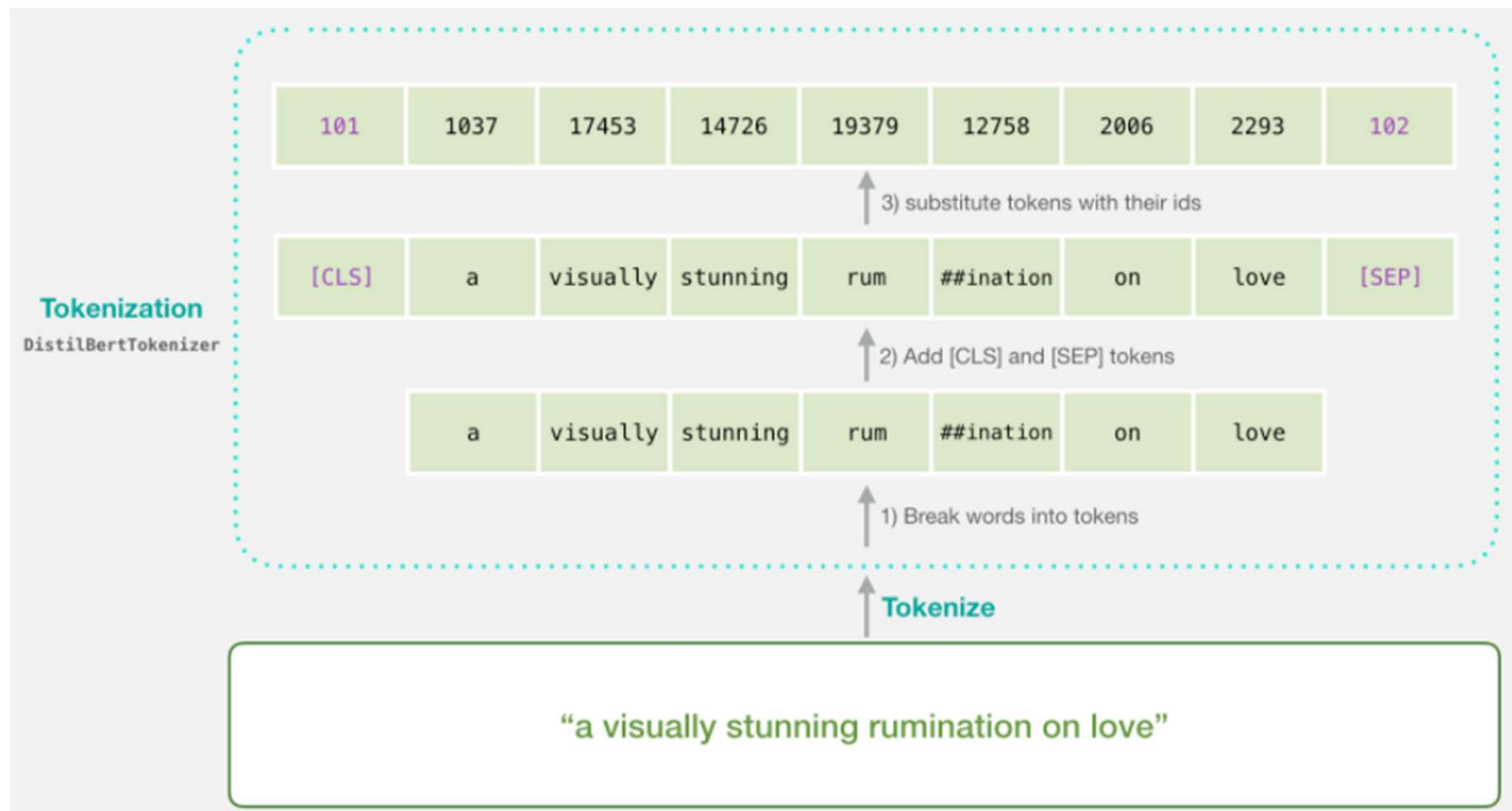


|       | Sentence Embeddings |         |     |       | label |
|-------|---------------------|---------|-----|-------|-------|
|       | 0                   | 1       | ... | 767   |       |
| 0     | -0.215              | -0.1402 | ... | 0.201 | 1     |
| 1     | -0.172              | -0.144  | ... | 0.371 | 0     |
| ...   | ...                 | ...     | ... | ...   | ...   |
| 1,999 | 0.124               | 0.014   | ... | 0.274 | 1     |

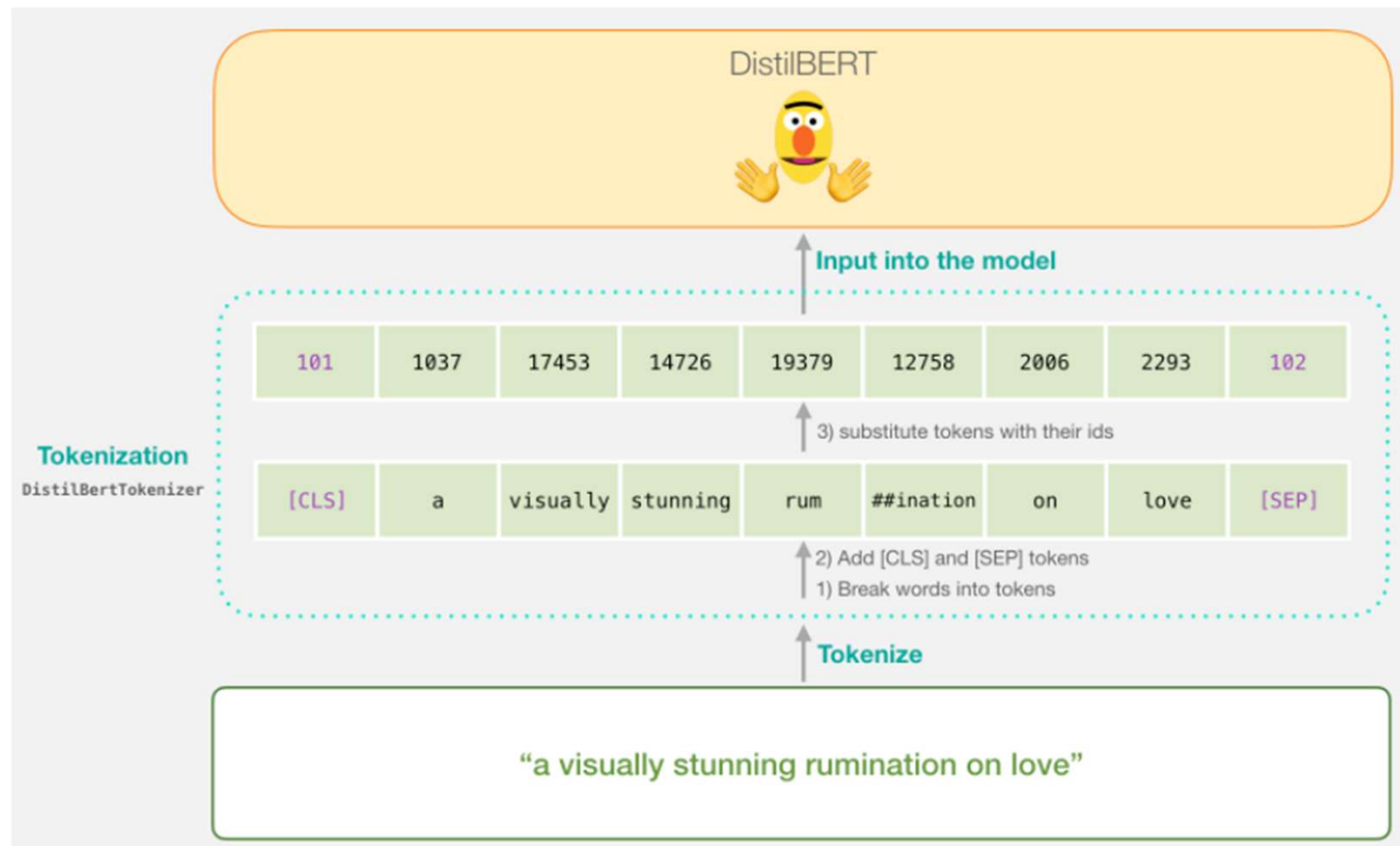
# Tokenization



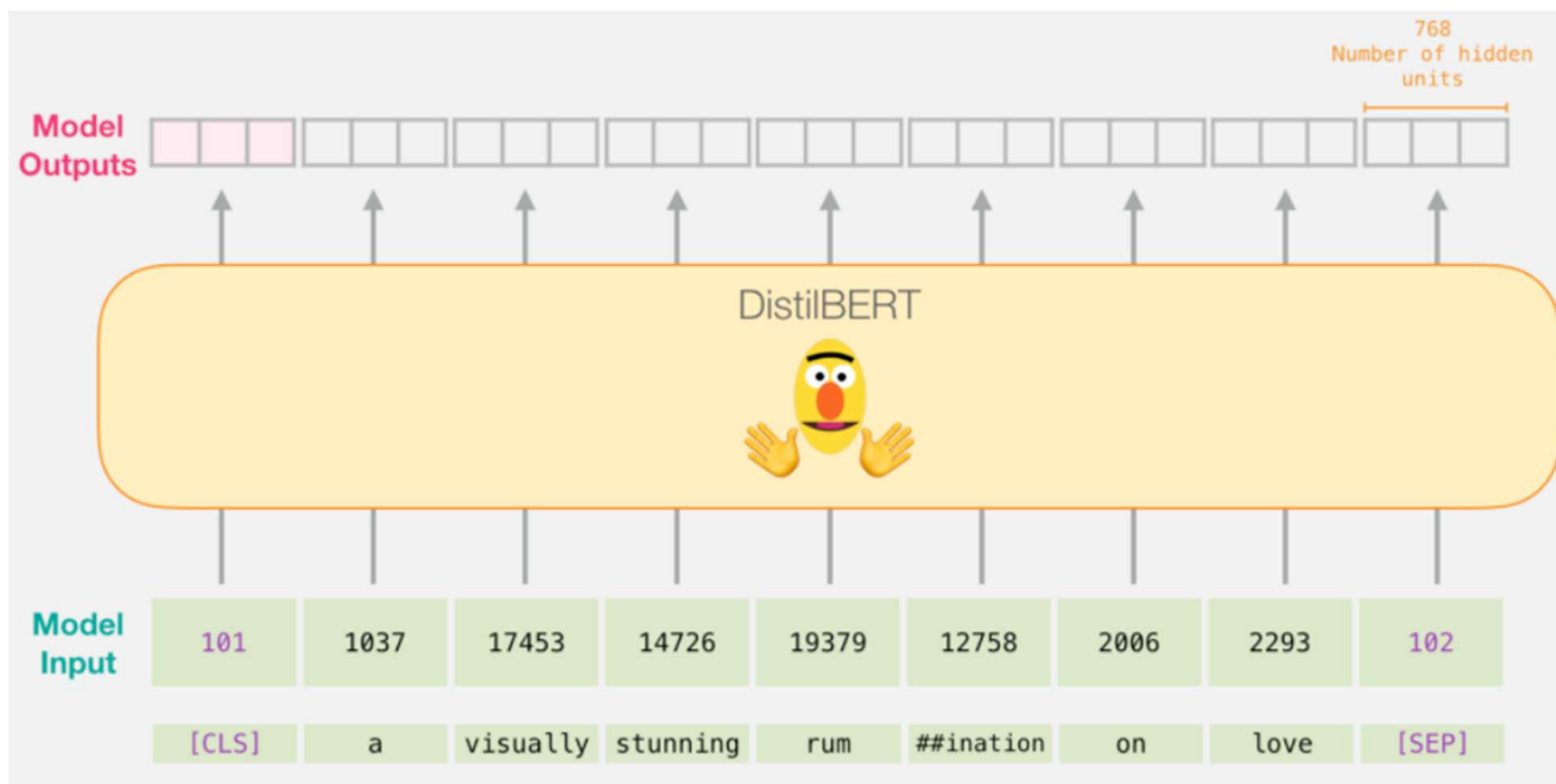
# Tokenization



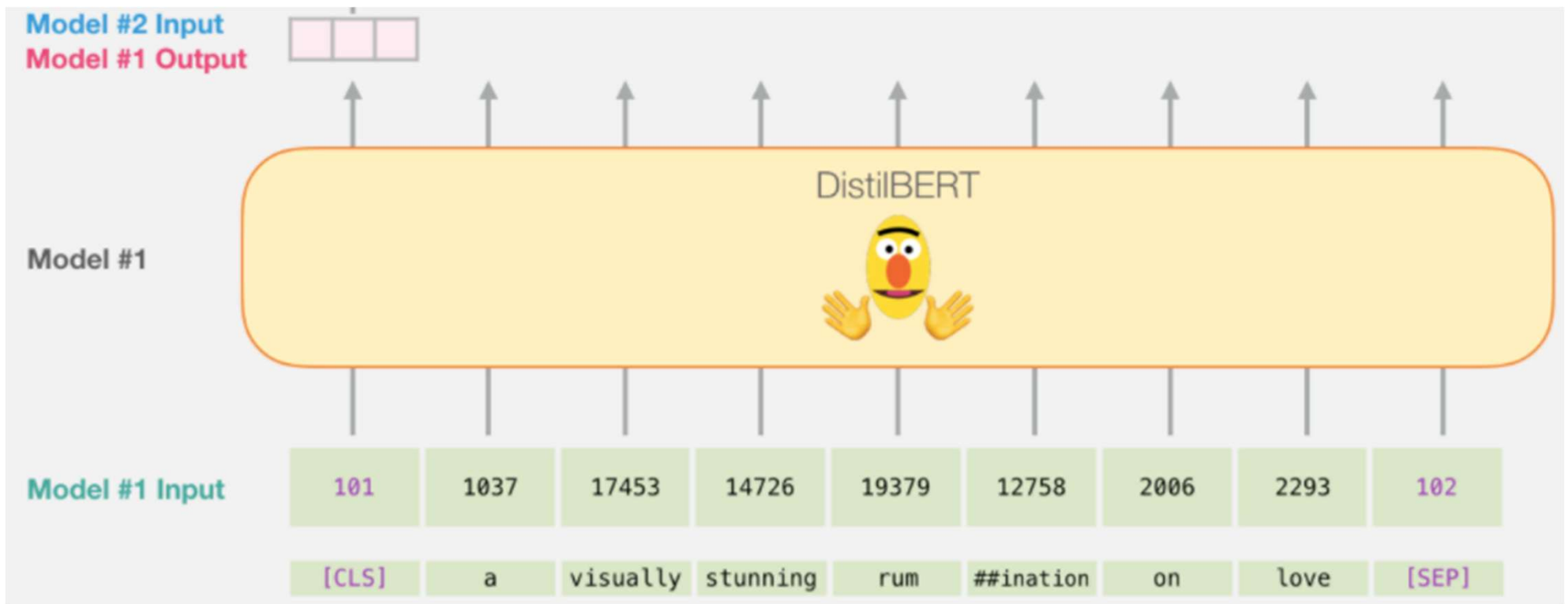
# Tokenization



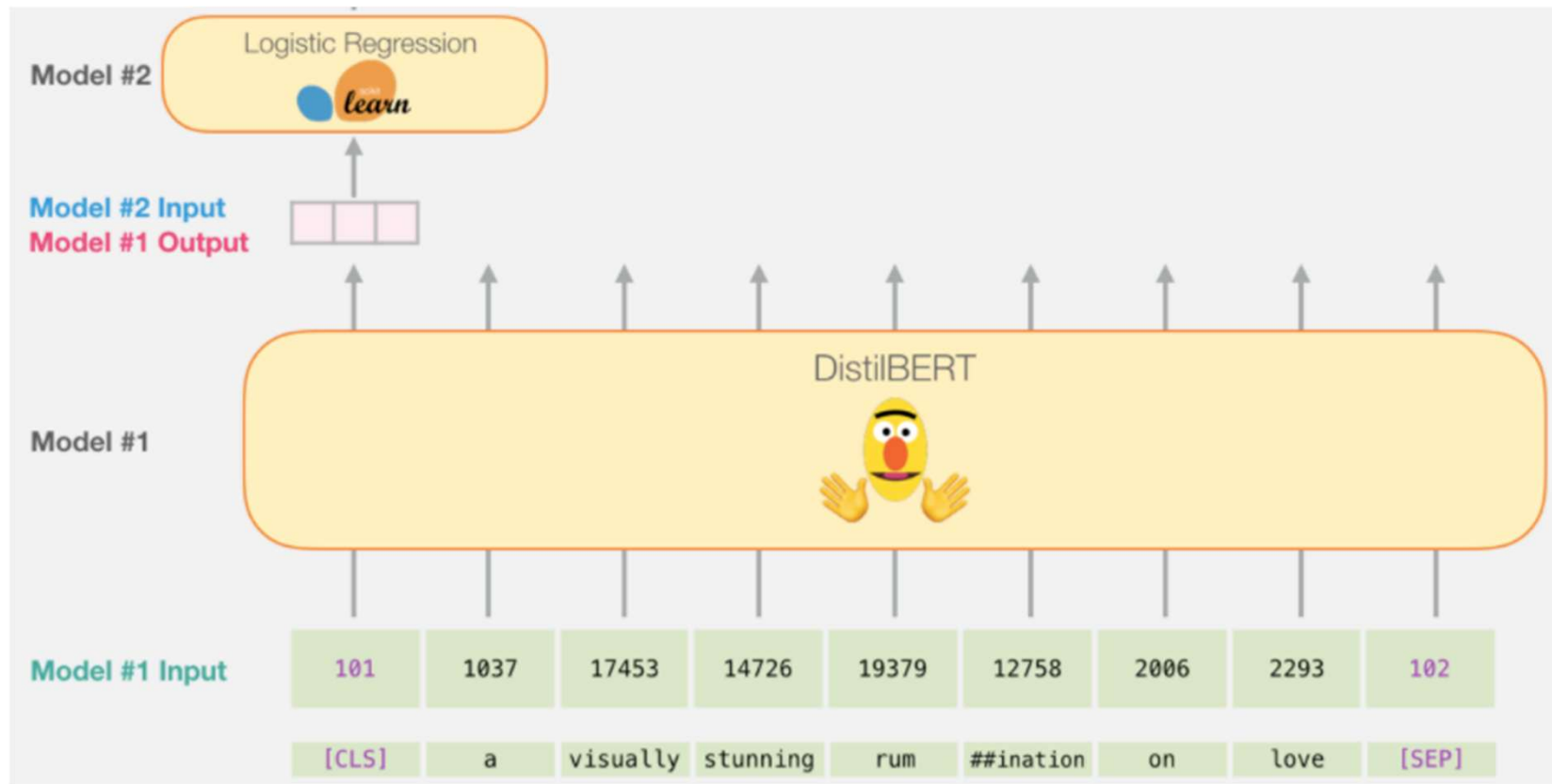
# Tokenization



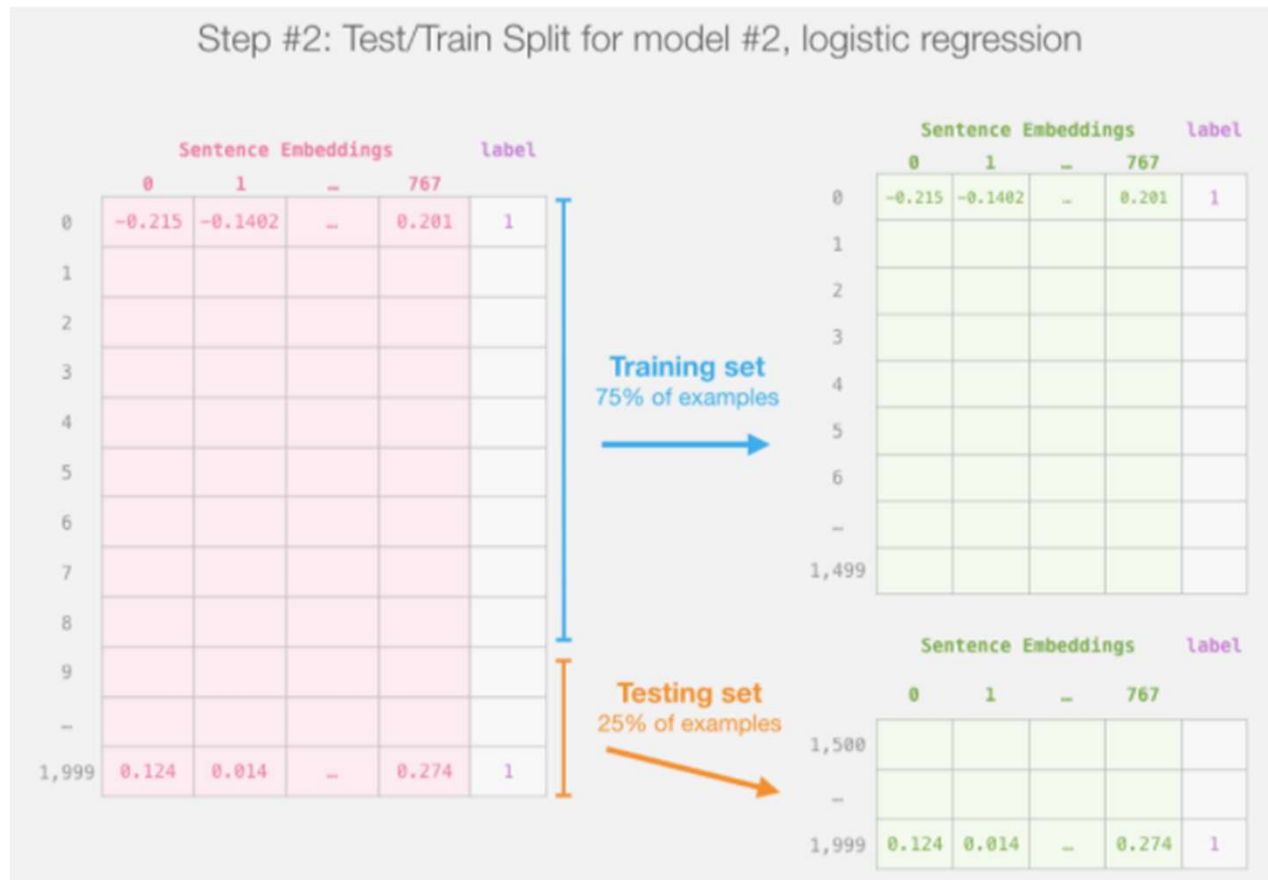
# Model 1 : [CLS] Token Embeddings



# Model 2: Linear Classifier



# Linear Classifier



# Linear Classifier

Step #3: Train the logistic regression model using the training set

|       | Sentence Embeddings |         |     |       | label |
|-------|---------------------|---------|-----|-------|-------|
|       | 0                   | 1       | ... | 767   |       |
| 0     | -0.215              | -0.1402 | ... | 0.201 | 1     |
| 1     |                     |         |     |       |       |
| 2     |                     |         |     |       |       |
| 3     |                     |         |     |       |       |
| 4     |                     |         |     |       |       |
| 5     |                     |         |     |       |       |
| 6     |                     |         |     |       |       |
| ...   |                     |         |     |       |       |
| 1,499 |                     |         |     |       |       |

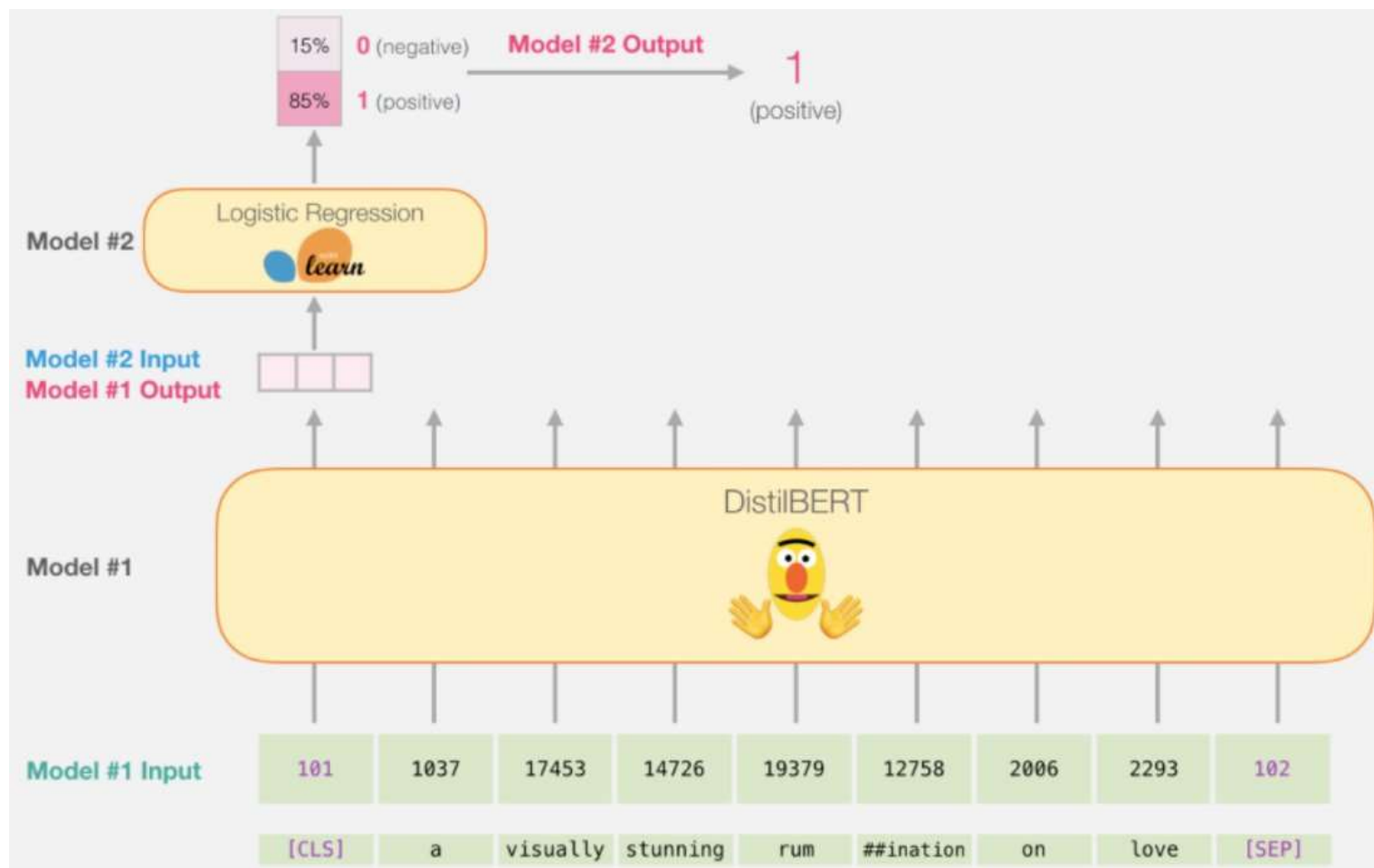
Model  
Training



Logistic  
Regression



# Review Classifier



# Further Reading

## Fastai

<https://course.fast.ai/>

## GPT-2

<https://transformer.huggingface.co/doc/gpt2-large>

## GPT-3

<https://openai.com/blog/gpt-3-apps/>

<https://arxiv.org/pdf/2005.14165.pdf>

<https://beta.openai.com/pricing>

## GPT-Neo

<https://www.eleuther.ai/>

